

MATH 301: Homework 3

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Sec 2.5 q2

If $S \subseteq \mathbb{R}, S \neq \emptyset$, show that S is bounded IFF $\exists I$, a closed bounded interval s.t. $S \subseteq I$

Consider the forward and reverse statements to prove:

$$\begin{cases} S \text{ is a bounded set} \implies \exists I \text{ (a closed bounded interval) } \ni S \subseteq I \\ \exists I \text{ (a closed bounded interval) } \ni S \subseteq I \implies S \text{ is a bounded set} \end{cases}$$

Suppose S is a bounded set

$$\implies \exists a = \inf S \quad b = \sup S$$

$$\implies \forall s \in S, (s \leq b) \wedge (s \geq a)$$

$$\therefore S \subseteq [a, b] \equiv S \subseteq I$$

Sec 2.5 q9

If $z \leq 0$ then $z \notin K_1$. If $w > 0$ then it follows from the Archimedian Property that there exists $n_w \in \mathbb{N}$ with $w \notin (n_w, \infty) = K_{n_w}$

Sec 3.1 q5.a

Use the definition of the limit of a sequence to establish the following limits:

$$\lim_{n \rightarrow \infty} f(n) = 0, \quad f(n) := \left(\frac{n}{n^2 + 1} \right)$$

Def (a.1): Limit Definition of Convergence

$$\lim_{n \rightarrow \infty} f(n) = 0 \iff \forall \epsilon > 0, \exists N(\epsilon) \in \mathbb{N} \ni \forall n \geq N(\epsilon), |x_n - L| < \epsilon$$

Backwards Search

Step	Reasoning / Simplification
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$\left \frac{n}{n^2+1} - 0 \right $	$< \epsilon$ Limit definition statement
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$\left \frac{n}{n^2+1} \right $	$< \epsilon$ Subtracting zero
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$\frac{ n }{n^2+1}$	$< \epsilon$ Property of absolute values
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$\frac{n}{n^2+1}$	$< \epsilon$ Since $n \in \mathbb{N}$, n is always positive
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$\frac{n}{n^2+1} < \frac{n}{n^2}$	$= \frac{1}{n}$ Bounding: $n^2 + 1 > n^2 \implies \frac{1}{n^2+1} < \frac{1}{n^2}$
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$\frac{1}{n}$	$< \epsilon$ Sufficient condition for the inequality
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n	$> \frac{1}{\epsilon} \quad \frac{1}{\epsilon} \in \mathbb{R}$
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$\uparrow$	By Archimedian Property
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Forward Logic: As $\frac{1}{\epsilon} \in \mathbb{R}$, this guarantees $\exists N(\epsilon) \in \mathbb{N} \ni N(\epsilon) > \frac{1}{\epsilon}$. This is the requirement to say $\forall n \geq N(\epsilon), n > \frac{1}{\epsilon}$. Therefore $\frac{1}{n} < \epsilon$. As $\frac{n}{n^2+1}$ we can generalize the apply the inequality for $f(n)$. Thus we can say $|f(n) - 0| < \epsilon$. This satisfies the epsilon definition of a limit convergence of a sequence $\therefore \lim_{n \rightarrow \infty} f(n) = 0$

Sec 3.1 q5.d

Use the definition of the limit of a sequence to establish the following limits:

$$\lim_{n \rightarrow \infty} f(n) = \frac{1}{2}, \quad f(n) := \left(\frac{n^2 - 1}{2n^2 + 3} \right)$$

$$\lim_{n \rightarrow \infty} f(n) = 1/2 \iff \forall \epsilon > 0, \exists N(\epsilon) \in \mathbb{N} \ni \forall n \geq N(\epsilon), |f(n) - L| < \epsilon$$

Backwards Search

Step	Reasoning / Simplification
$\left \frac{n^2-1}{2n^2+3} - \frac{1}{2} \right < \epsilon$	Formal definition of a limit: $ f(n) - L < \epsilon$
$\left \frac{2(n^2-1)-(2n^2+3)}{2(2n^2+3)} \right < \epsilon$	Find a common denominator: $2(2n^2 + 3)$
$\left \frac{2n^2-2-2n^2-3}{4n^2+6} \right < \epsilon$	Distribute and simplify the numerator
$\left \frac{-5}{4n^2+6} \right < \epsilon$	Combine like terms
$\frac{5}{4n^2+6} < \epsilon$	Apply absolute value: $ -5 = 5$ and $4n^2 + 6 > 0$
$\frac{5}{4n^2+6} < \frac{5}{4n^2} < \epsilon$	Bounding: $4n^2 + 6 > 4n^2 \implies$ fraction is smaller
$\frac{5}{4n^2} \leq \frac{5}{4n} < \epsilon$	Since $n \geq 1, n^2 \geq n$, so $\frac{1}{n^2} \leq \frac{1}{n}$
$\frac{5}{4n} < \epsilon$	Sufficient condition to satisfy original inequality
$\frac{1}{n} < \frac{4\epsilon}{5}$	Multiply by $\frac{4}{5}$ on both sides
$n > \frac{5}{4\epsilon}$	Take the reciprocal (flips the inequality sign)

↑ Choose $N > \frac{5}{4\epsilon}$ by the Archimedean Property

Forward Logic: Let $\epsilon > 0$ be given. As $\frac{5}{4\epsilon} \in \mathbb{R}$, the Archimedean Property guarantees there exists a natural number $N(\epsilon) \in \mathbb{N}$ such that: $N(\epsilon) > \frac{5}{4\epsilon}$. This is the requirement to say that for all $n \geq N(\epsilon)$, the inequality $n > \frac{5}{4\epsilon}$ holds. By taking the reciprocal and multiplying, we obtain: $\frac{5}{4n} < \epsilon$. Since $n \geq 1$, we know $n^2 \geq n$, which implies $\frac{1}{n^2} \leq \frac{1}{n}$. Thus, we can apply the following chain of inequalities to $f(n)$: $\left| \frac{n^2-1}{2n^2+3} - \frac{1}{2} \right| = \frac{5}{4n^2+6} < \frac{5}{4n^2} \leq \frac{5}{4n}$. Because we established $\frac{5}{4n} < \epsilon$, it follows by transitivity that: $|f(n) - \frac{1}{2}| < \epsilon$. This satisfies the epsilon definition of a limit for the convergence of a sequence. $\therefore \lim_{n \rightarrow \infty} \frac{n^2-1}{2n^2+3} = \frac{1}{2}$

Sec 3.1 q8

Prove that $\lim(x_n) = 0 \iff \lim(|x_n|) = 0$. Give an example to show that the convergence of $(|x_n|)$ need not imply the convergence of (x_n) .

Lemma (a.1) : Absolute Value Properties

$$\forall a \in \mathbb{R}, |a| \implies \begin{cases} a \in \mathbb{P} \\ -a \in \mathbb{P} \\ a = 0 \end{cases}$$

$$\therefore |x_n| = (x_n \in \mathbb{P}) \oplus (-x_n \in \mathbb{P}) \oplus (x_n \in \{0\})$$

Lemma (a.2) : Absolute Value Properties

$$\text{Consider } ||x_n| - 0| = \begin{cases} (x_n \in \mathbb{P}) & \implies & ||x_n| - 0| = |x_n - 0| \\ (-x_n \in \mathbb{P}) & \implies & ||x_n| - 0| = | -(-x_n) - 0| = |x_n - 0| \\ (x_n \in \{0\}) & \implies & ||x_n| - 0| = |x_n - 0| \end{cases}$$

$$\therefore ||x_n| - 0| = |x_n - 0|$$

Case $(\rightarrow)$: $\lim(x_n) = 0 \rightarrow \lim(|x_n|) = 0$

Observe $||x_n| - 0| < \epsilon$ matches *Def (a.1)*. Naturally, should the inequality $||x_n| - 0| = |x_n - 0|$ then this would imply their limits are the same as defined prior. This equality is given via *Lemma (a.2)*.

$$\therefore \lim(x_n) = 0 \rightarrow \lim(|x_n|) = 0$$

Case $(\leftarrow)$: $\lim(x_n) = 0 \leftarrow \lim(|x_n|) = 0$

Applying the same *Def (a.1)*, *Lemma (a.2)* we arrive to the same conclusion following the same logic.

$$\therefore \lim(x_n) = 0 \leftarrow \lim(|x_n|) = 0$$

Ex. $\lim(|x_n|)$ converges $\nrightarrow \lim(x_n)$

Lemma (a.3) : $|(-1)^n| = 1$

$$P(n) : |(-1)^n| = 1$$

$$P(1) : |-1| = 1$$

Suppose $P(k) : |(-1)^k| = 1$ holds.

$$|(-1)^k| |(-1)| = 1 |(-1)|$$

$$P(k+1) : |(-1)^{k+1}| = 1$$

As $P(k) \implies P(k+1) \therefore$ by PMI $P(n)$ holds, $\forall n \in \mathbb{N}$

For the sake of contradiction, suppose $x_n := (-1)^n$ and $\lim(|x_n|)$ converges $\rightarrow \lim(x_n)$

$X := \{|(-1)^n| : n \in \mathbb{N}\} = \{1_1, 1_2, \dots, 1_n\}$ By *Lemma (a.3)*

The sequence never changes thus, is convergent on 1

Note: $x_n = (-1)^n$ however is monotonic thus does not converge $\nexists$

Sec 3.1 q9

Show that $\forall n \in \mathbb{N}, (x_n \geq 0) \wedge (\lim_{n \rightarrow \infty} x_n = 0) \implies \lim_{n \rightarrow \infty} \sqrt{x_n} = 0$

Suppose $\forall n \in \mathbb{N}, (x_n \geq 0) \wedge (\lim_{n \rightarrow \infty} x_n = 0)$

$\lim_{n \rightarrow \infty} x_n = 0 \implies \exists n_\epsilon \text{ s.t. } \forall n \geq n_\epsilon, |x_n - 0| < \epsilon^2, (0 < \epsilon)$

$$0 < x_n < \epsilon^2 \iff 0 < \sqrt{x_n} < \epsilon$$

$\therefore \forall n \in \mathbb{N}, (x_n \geq 0) \wedge (\lim_{n \rightarrow \infty} x_n = 0) \implies \lim_{n \rightarrow \infty} \sqrt{x_n} = 0$

Sec 3.1 q14

Prove: $b \in \mathbb{R}, 0 < b < 1 \implies \lim_{n \rightarrow \infty} nb^n = 0$

Theorem (a.1): Binomial Theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Expanded form:

$$(a + b)^n = \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n} b^n$$

Example for $n = 3$:

$$(a + b)^3 = \binom{3}{0} a^3 + \binom{3}{1} a^2 b + \binom{3}{2} a b^2 + \binom{3}{3} b^3$$

$$= a^3 + 3a^2 b + 3ab^2 + b^3$$

$$b \in \mathbb{R}, 0 < b < 1 \implies \exists 0 < c \in \mathbb{R} \ni b = \frac{1}{1+c}$$

$$\implies b^n = \frac{1}{1+c^n}$$

$$\frac{1}{1+c^n} < \frac{n}{1+c^n}$$

$$\lim_{n \rightarrow \infty} \frac{n}{1+c^n} = 0 \implies \lim_{n \rightarrow \infty} \frac{1}{1+c^n} = 0$$

$$(1+c)^n = \sum_{k=1}^{\infty} \binom{n}{k} c^k \quad \text{By Theorem (a.1)}$$

Sec 3.2 q2

Give an example of two divergent sequences X, Y such that:

- Their sum $X + Y$ converges
- Their product XY converges

$$X := (-1)^n \quad Y = (-1)^{n+1}$$

$$X = (-1)^n \implies \begin{cases} k \in \mathbb{N} \cup \{0\}, n = 2k & X' = \frac{1}{n} \\ k \in \mathbb{N} \cup \{0\}, n = 2k+1 & X'' = \frac{-1}{n} \end{cases}$$

$$\lim_{n \rightarrow \infty} X' = 0 = X''$$

$$x'' \in X'' \implies x'' \in [-1, 0)$$

$$x' \in X' \implies x' \in (0, 1]$$

$$\nexists a \in \mathbb{R} \ni a \in [-1, 0) \wedge a \in (0, 1]$$

$\therefore X$ does not converge

Sec 3.2 q4

Show that if X, Y s.t. X converges to $x \neq 0$ and XY converges, then Y converges.

$$\text{Given: } \lim_{n \rightarrow \infty} X = L_x \neq 0 \implies \forall \epsilon > 0, \exists n_\epsilon \ni \forall n \geq n_\epsilon, |x_n - L_x| < \epsilon$$

$$\text{Given: } \lim_{n \rightarrow \infty} XY = L_a \neq 0 \implies \forall \epsilon > 0, \exists n_\epsilon \ni \forall n \geq n_\epsilon, |x_n y_n - L_a| < \epsilon$$

$$\lim_{n \rightarrow \infty} Y = \frac{\lim_{n \rightarrow \infty} XY}{\lim_{n \rightarrow \infty} X} = \frac{L_a}{L_x}$$

Sec 3.2 q6.b

Find the limit of: $\lim \left(\frac{(-1)^n}{n+2} \right)$

$$\frac{(-1)^n}{n+2} \implies \begin{cases} \lim(\frac{-1}{n+2}) = a \\ \lim(\frac{1}{n+2}) = b \end{cases}$$
$$a = b \implies \lim \left(\frac{(-1)^n}{n+2} = a = b \right)$$
$$\frac{|1|}{n+2} = \frac{1}{n+2} < \frac{1}{n} < \epsilon$$
$$\therefore \lim \left(\frac{(-1)^n}{n+2} \right) = 0$$

Sec 3.2 q6.c

Find the limit of: $\lim \left(\frac{\sqrt{n}-1}{\sqrt{n}+1} \right)$

$$\frac{\sqrt{n}-1}{\sqrt{n}+1}$$

$$\frac{\sqrt{n}-1}{\sqrt{n}+1} \cdot \frac{\frac{1}{\sqrt{n}}}{\frac{1}{\sqrt{n}}}$$

$$\frac{\frac{\sqrt{n}}{\sqrt{n}} - \frac{1}{\sqrt{n}}}{\frac{\sqrt{n}}{\sqrt{n}} + \frac{1}{\sqrt{n}}}$$

$$\frac{1 - \frac{1}{\sqrt{n}}}{1 + \frac{1}{\sqrt{n}}}$$

$$\lim_{n \rightarrow \infty} \frac{1-0}{1+0} = 1$$

Sec 3.2 Q9

Let $y_n := \sqrt{n+1} - \sqrt{n}$ for $n \in \mathbb{N}$. Show that $\sqrt{n}y_n$ converges. Find the limit.

y_n	$= \sqrt{n+1} - \sqrt{n}$	Given
	$= (\sqrt{n+1} - \sqrt{n}) \cdot \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}}$	Multiply by conjugate
	$= \frac{(n+1)-n}{\sqrt{n+1} + \sqrt{n}}$	Difference of squares
	$= \frac{1}{\sqrt{n+1} + \sqrt{n}}$	Simplify
$\sqrt{n}y_n$	$= \frac{\sqrt{n}}{\sqrt{n+1} + \sqrt{n}}$	Multiply y_n by $\sqrt{n}$
	$= \frac{\sqrt{n}}{\sqrt{n+1} + \sqrt{n}} \cdot \frac{\frac{1}{\sqrt{n}}}{\frac{1}{\sqrt{n}}}$	Multiply by 1 to simplify
	$= \frac{1}{\frac{\sqrt{n+1}}{\sqrt{n}} + 1}$	Distribute $\frac{1}{\sqrt{n}}$
	$= \frac{1}{\sqrt{\frac{n+1}{n}} + 1}$	Combine square roots
	$= \frac{1}{\sqrt{1 + \frac{1}{n}} + 1}$	Simplify fraction
$\lim_{n \rightarrow \infty} \sqrt{n}y_n$	$= \frac{1}{\sqrt{1+0} + 1}$	Apply limit (since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$)
	$= \frac{1}{2}$	Final Answer